

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Geology & Geophysics

2. **Subject Name:** Engineering Geology Lab. **L-T-P:** 0-0-3 **Credits:** 2

3. **Pre-requisites:** NA

4. **Syllabus and reference books:**

Syllabus:

Solving rock engineering problems using Mohr circle and Mohr-Coulomb failure envelope, Application of Griffith failure criterion, Solving borehole-fracture problems using hemispherical projections, Characterization of rock masses using rock mass classifications, Slope failure mechanisms and kinematic analysis, Solving problems related to soil tests, Demonstration of various rock mechanics and index tests, Calculation and analysis of determined rock engineering parameters.

Reference Books:

- 1) "Practical Engineering Geology" by Steve Hencher
- 2) "Introduction to Rock Mechanics" by R. E. Goodman
- 3) "Geology for Ground Engineering Projects" by Chris J. N. Fletcher
- 4) "Soil Mechanics and Foundations" by Muni Budhu
- 5) "Engineering Geology" by F. G. Bell
- 6) "Engineering in Rocks for Slopes, Foundations and Tunnels" Edited by T. Ramamurthy
- 7) "A Geology for Engineers" by F. G. H. Blyth and M. H. de Freitas

5. **Class-wise break-up:**

Sl. No.	Topic	No. of lectures
1	Solving rock engineering problems using failure criteria	6
2	Solving borehole-fracture problems using hemispherical projections	6
3	Rock mass classifications	3
4	Slope failure mechanisms and kinematic analysis	6
5	Solving problems related to soil tests	6
6	Demonstration of various rock mechanics and index tests, Calculation and analysis of determined rock engineering parameters.	9
Total number of hours		36